Doc: ET 28 (6756)

BUREAU OF INDIAN STANDARDS
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Draft *Indian Standard*

PHOTOVOLTAIC (PV) ARRAYS – DESIGN REQUIREMENTS

Last date for receipt of comments is: **25-09-2014**

Solar Photovoltaic Energy Systems Sectional Committee, ET 28

NATIONAL FOREWORD

This draft Indian Standard which is identical with IEC/TS 62548 : 2013 'Photovoltaic (PV) arrays – Design requirements' issued by the International Electrotechnical Commission (IEC) would be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are however not identical to those used in Indian Standards. Attention is particularly drawn to the following:

a) Wherever the words 'International Standard' appear referring to this standard, they should be read as 'Indian Standard'.

b) Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standards which are to be substituted in their places, are listed below along with its degree of equivalence for the edition indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60228 : 2004 Conductors of insulated cables	IS 8130:2013 Conductors for insulated electric cables and flexible cords (second revision)	Identical
IEC 60269-6 Low-voltage fuses – Part 6: Supplementary requirements for fuse-links for the protection of solar photovoltaic energy systems	DOC.ETD 39(6430) Low-voltage fuses – Part 6: Supplementary requirements for fuse-links for the protection of solar photovoltaic energy systems	Identical with IEC 60269-6 : 2010

IEC 60364 (all parts) Low-voltage electrical installations	SP 30 : 2011 National Electrical Code (first revision)	Technically Equivalent
IEC 60364-4-41:2005 Low-voltage electrical installations – Part 4-41: Protection for safety –Protection against electric shock	SP 30 : 2011 National Electrical Code (first revision)	do
IEC 60364-5-54:2011 Low-voltage electrical installations – Part 5-54: Selection and erection of electrical equipment – Earthing arrangements and protective conductors	SP 30 : 2011 National Electrical Code (first revision)	do
IEC 60364-7-712:2002 Electrical installations of buildings – Part 7-712: Requirements forspecial installations or locations – Solar photovoltaic (PV) power supply systems	SP 30 : 2011 National Electrical Code (first revision)	Technically Equivalent
IEC 60529 Degrees of protection provided by enclosures (IP Code)	IS/IEC 60529 : 2001 Classification of Degrees of protection provided by enclosures (IP Code)	Identical
IEC 60898-2 Circuit-breakers for over current protection for household and similar installations – Part 2: Circuit-breakers for a.c. and d.c. operation	IS 60898(Part 2) : 2003 Electrical Accessories - circuit breakers for over protection for household and similar installations Part 2 Circuit breakers for a.c. and d.c. operation	Identical with IEC 60898-2 : 2003
IEC 60947-1 Low-voltage switchgear and controlgear – Part 1: General rules	IS 12762 (Part 1) : 2010 Photovoltaic devices: Part 1 Measurement of photovoltaic current voltage characteristics	Identical with IEC 60904-1 : 2006
IEC 60947-2 Low-voltage switchgear and controlgear – Part 2: Circuit breakers	IS 12762 (Part 2) : 2013 Photovoltaic devices: Part 2 Requirements for reference solar cells	Identical with IEC 60904-2 : 2007
IEC 60947-3 Low-voltage switchgear and controlgear – Part 3: Switches, disconnectors, switch-disconnectors and fuse-combination units	IS 12762 (Part 3) : 2013 Photovoltaic devices: Part 3 Measurement principles for terrestrial photovoltaic (PV) solar devices with reference spectral irradiance data (First Revision)	Identical with IEC 60904-3 : 2008
IEC 61215 : 2005 Crystalline silicon terrestrial photovoltaic (PV) modules − Designqualification and type approval	IS 14286 : 2010 Crystalline silicon terrestrial photovoltaic (PV) modules - Design qualiication and type approval (first revision)	Identical with IEC 61215 : 2005

IEC 61646 Thin-film terrestrial photovoltaic (PV) modules − Design qualification and typeapproval	IS 16077 : 2013 Thin-film terrestrial photovoltaic (PV) modules Design qualification and type approval	Identical with IEC 61646 : 2008
IEC 61730-1 : 2004 Photovoltaic (PV) module safety qualification − Part 1: Requirements for construction	IS 61730 (Part 1) : 2004 Photovoltaic (PV) module safety qualification:Part 1 Requirments for construction	Identical with IEC 61730-1 : 2004
IEC 61730-2 : 2004 Photovoltaic (PV) module safety qualification − Part 2: Requirements for testing	IS 61730 (Part 2) : 2004 Photovoltaic (PV) module safety qualification : Part 2 Requirements for testing	Identical with IEC 61730-2 : 2004
IEC 62109-1 : 2010 Safety of power converters for use in photovoltaic power systems – Part 1:General requirements	Doc : ET 28(6417) Safety of power converters for use in photovoltaic power systems – Part 1:General requirements (Under print)	Identical with IEC 62109-1 : 2010
IEC 62109-2 Safety of power converters for use in photovoltaic power systems – Part 2:Particular requirements for inverters	Doc : ET 28(6418) Safety of power converters for use in photovoltaic power systems – Part 1:Particular requirements for inverters (Under print)	Identical with IEC 62109-2 : 2011

The technical committee has reviewed the provisions of the following International Standard referred in this adopted standard and has decided that it is acceptable for use in conjunction with this standard:

International Standards	*Title*
IEC 60287 (all parts)	Electric cables – Calculation of the current rating
IEC 60332-1-2:2004	Tests on electric and optical fibre cables under fire conditions – Part 1-2: Test for vertical flame propagation for a single insulated wire or cable – Procedure for 1 kWpre-mixed flame
IEC 62305-2	Protection against lightning – Part 2: Risk management
IEC 62305-3	Protection against lightning – Part 3: Physical damage to structures and life hazard
IEC 62305-4	Protection against lightning – Part 4: Electrical and electronic systems within structures
IEC 62446	Grid connected photovoltaic systems – Minimum requirements for system documentation, commissioning tests and inspection
EN 50521	Connectors for photovoltaic systems – Safety requirements and tests

Only the English language text of the International Standard has been retained while adopting it as an Indian Standard, and as such the page numbers given here are not the same as in IEC Publication.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note : The technical content of the document is not available on website. For details, please refer the corresponding IEC/TS 62548 : 2013 . Or kindly contact :

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Doc: ET 28 (6757)

BUREAU OF INDIAN STANDARDS
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Draft *Indian Standard*

PHOTOVOLTAIC CONCENTRATORS (CPV) – PERFORMANCE TESTING – PART 1: STANDARD CONDITIONS

Last date for receipt of comments is: **25-09-2014**

Solar Photovoltaic Energy Systems Sectional Committee, ET 28

NATIONAL FOREWORD

This draft Indian Standard (Part 1) which is identical with IEC 62670-1 : 2013 'Photovoltaic concentrators (CPV) – Performance testing – Part 1: Standard conditions' issued by the International Electrotechnical Commission (IEC) was adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council.

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

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b) Comma (,) has been used as a decimal marker while in Indian Standards, the current practice is to use a point (.) as the decimal marker.

In this adopted standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standard which is to be substituted in its respective place is listed below along with its degree of equivalence for the edition indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 60904-3 Photovoltaic devices – Part 3: Measurement principles for terrestrial photovoltaic (PV) solar devices with reference spectral irradiance data	IS 12762 (Part 3) : 2013 Photovoltaic devices: Part 3 Measurement principles for terrestrial photovoltaic (PV) solar devices with reference spectral irradiance data (first revision)	Identical

Only the English language text of the International Standard has been retained while adopting it as an Indian Standard, and as such the page numbers given here are not the same as in IEC Publication.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note : The technical content of the document is not available on website. For details, please refer the corresponding IEC 62670-1 : 2013 . Or kindly contact :

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Doc: ET 28 (6758)

BUREAU OF INDIAN STANDARDS
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Draft *Indian Standard*
PHOTOVOLTAIC (PV) MODULES – AMMONIA CORROSION TESTING

Last date for receipt of comments is: **25-09-2014**

Solar Photovoltaic Energy Systems Sectional Committee, ET 28

NATIONAL FOREWORD

This draft Indian Standard which is identical with IEC 62716 : 2013 'Photovoltaic (PV) modules – Ammonia corrosion testing' issued by the International Electrotechnical Commission (IEC) would be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

The text of IEC Standard has been approved as suitable for publication as an Indian Standard without deviations. Certain conventions are, however, not identical to those used in Indian Standards. Attention is particularly drawn to the following:

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In this adopted standard, references appear to certain International Standards for which Indian Standards also exist. The corresponding Indian Standard which is to be substituted in its respective place is listed below along with its degree of equivalence for the edition indicated:

International Standard	*Corresponding Indian Standard*	*Degree of Equivalence*
IEC 61215:2005, Crystalline silicon terrestrial photovoltaic (PV) modules − Designqualification and type approval	IS 14286 : 2010 Crystalline silicon terrestrial photovoltaic (PV) modules - Design qualiication and type approval (first revision)	Identical to IEC 61215 : 2005
IEC 61646, Thin-film terrestrial photovoltaic (PV) modules − Design qualification and typeapproval	IS 16077 : 2013 Thin-film terrestrial photovoltaic (PV) modules Design qualification and type approval	Identical to IEC 61646 : 2008
IEC 61730-2:2004, Photovoltaic (PV) module safety qualification − Part 2: Requirements for testing	IS 61730 (Part 2) : 2004 Photovoltaic (PV) module safety qualification : Part 2 Requirements for testing	Identical to IEC 61730-2 : 2004

ISO/IEC 17025, General requirements for the competence of testing and calibration laboratories	IS/ISO/IEC 17025 : 2005 General requirements for competence of testing and calibration laboratories	Identical

The technical committee has reviewed the provisions of the following International Standard referred in this adopted standard and has decided that it is acceptable for use in conjunction with this standard:

International Standards	*Title*
ISO 6988:1985	Metallic and other non organic coatings – Sulfur dioxide test with general condensation of moisture

Only the English language text of the International Standard has been retained while adopting it as an Indian Standard, and as such the page numbers given here are not the same as in IEC Publication.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

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Doc: ET 28 (6759)

BUREAU OF INDIAN STANDARDS
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Draft *Indian Standard*

PHOTOVOLTAIC SYSTEMS – SPECIFICATIONS FOR SOLAR TRACKERS

Last date for receipt of comments is: **25-09-2014**

Solar Photovoltaic Energy Systems Sectional Committee, ETD 28

NATIONAL FOREWORD

This draft Indian Standard which is identical with IEC/TS 62727 : 2012 'Photovoltaic systems – Specifications for solar trackers' issued by the International Electrotechnical Commission (IEC) would be adopted by the Bureau of Indian Standards, on the recommendation of the Solar Photovoltaic Energy Systems Sectional Committee and approval of the Electrotechnical Division Council

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For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated expressing the result of a test, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Note : The technical content of the document is not available on website. For details, please refer the corresponding IEC/TS 62727 : 2012 . Or kindly contact :

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